

# CALLUSES AND CORNS

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**Pharm***Achieve*

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# DISCLAIMER

- While progressing through this learning and using any of the resources provided, please keep in mind:
  - ✓ the scope of practice for pharmacists within your province or territory
  - ✓ the policies and procedures of your designated place of work
  - ✓ your scope of practice may be more limited or extensive than presented in this eLearning module
- The treatment examples provided are intended to illustrate possible indicated options for managing the specified common/minor ailment or condition. Pharmacists should:
  - ✓ follow provincial/territorial regulations regarding treatment options they are authorized to prescribe
  - ✓ exercise professional judgment at all times to prescribe the most appropriate option for each patient
  - ✓ refer to the latest clinical guidelines and product monographs

# LEARNING OBJECTIVES

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## CALLUSES AND CORN



Describe the pathophysiology and clinical presentation of calluses and corns



Recognize the possible risk factors and causes of calluses and corns



Understand how to assess calluses and corns



Identify red flags for referral and differential diagnoses



Outline non-pharmacological and pharmacological treatment options for calluses and corns



Identify side effects and contraindications to specific treatments



Understand monitoring parameters for the treatment of calluses and corns



# LEGEND

**ABCDE:** Asymmetry, Borders, Colour, Diameter, and Evolution

**PVD:** Peripheral Vascular Disease

**SCHOLAR/HAMS:** Symptoms, Characteristics, History, Onset, Location, Aggravating factors and Remitting factors, Health conditions, Allergies, Medications, Social history

A top-down view of a doctor's desk. A person in a white lab coat is typing on a silver laptop. To the right, a blue stethoscope rests on a white clipboard with a blue folder underneath. A smartphone lies on the desk next to the laptop. In the foreground, a silver keyboard is visible. The desk is covered with various medical and office supplies, including a pen, glasses, and a small container. The background is a light-colored wooden surface. The text 'CLINICAL PRESENTATION & ASSESSMENT' is overlaid on the left side of the image.

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# CLINICAL PRESENTATION & ASSESSMENT



# CLINICAL PRESENTATION

## CORNS

- Layers of compacted skin which may have a central core (radix) over a bony prominence
- May be yellow in appearance
- Underlying bursitis and nerve irritation causes pain

## HARD CORN

Tough layers of skin often on knuckles of toes or soles of the feet

## SOFT CORN

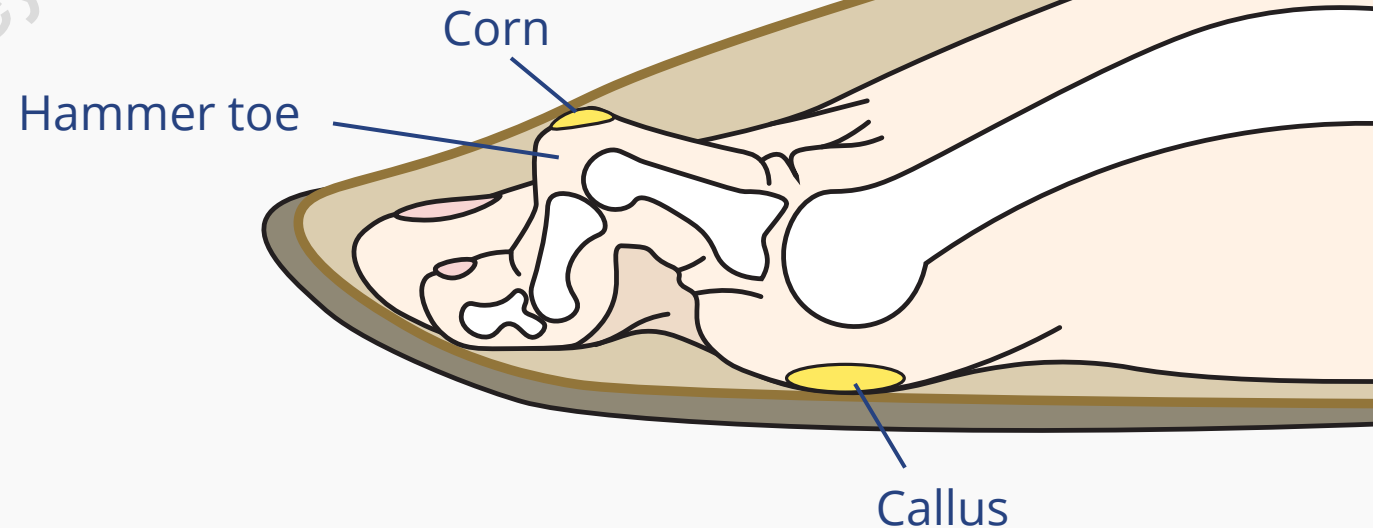
- Previously a hard corn that absorbed moisture and has softened
- Mostly located between toes



# CLINICAL PRESENTATION

## CALLUS

- Buildup of keratinized epidermis from repeated pressure or friction
- Even thickness with the absence of a central core (radix)
- Typically not painful
- Tend to occur on ball or heel of foot



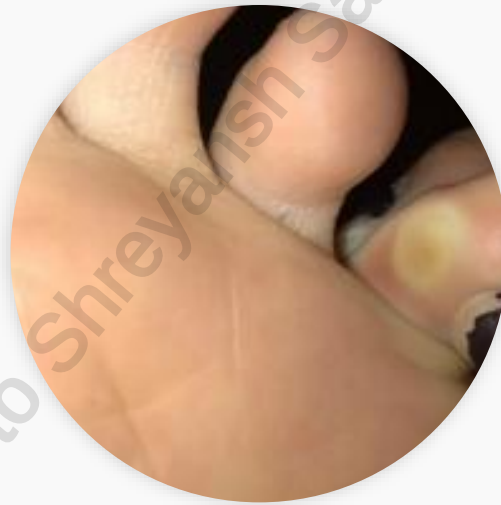


# CLINICAL PRESENTATION

**HARD CORN**



**SOFT CORN**



**CALLUS**





# PATHOPHYSIOLOGY

Excessive friction  
and/or pressure on skin



Increased keratinization  
of epidermis



Callus/corn  
formation

Repeated pressure  
and/or friction of skin  
over bones in the feet

In order to protect the  
skin from breakdown,  
the body increases  
keratotic tissue

Buildup of keratinized  
tissue leads to corns or  
calluses



# RISK FACTORS

- Chronic excessive pressure or friction on areas of feet
- Abnormal gait
- Ill-fitting footwear
  - High-heeled or tight-fitting shoes, internal stitching or seams
  - Wearing shoes without socks
- Abnormal foot mechanics or physical deformities
  - Bunions, hammertoes, clubfoot, mallet toes, or bumps that may develop due to rheumatoid arthritis
- Weight gain
- High levels of activity
- Repeated, increased pressure points on the feet



# DIFFERENTIAL DIAGNOSES

| CONDITION                    | PRESENTATION                                                                                                                                                                                                                    |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Plantar warts                | <ul style="list-style-type: none"><li>• Hard and flat with elevated rough surfaces</li><li>• Pinpoint bleeding upon excision</li></ul>                                                                                          |
| Onychomycosis                | <ul style="list-style-type: none"><li>• Fungal infection of the nail and/or nail bed</li><li>• Discolouration of nail and nail bed (white, yellow, green, or black)</li><li>• Dry or scaly skin around a brittle nail</li></ul> |
| Bunion                       | <ul style="list-style-type: none"><li>• Persistent or intermittent pain, inflammation, redness</li><li>• Bony protrusion on joint of large toe</li><li>• Bunionette if located on pinky toe</li></ul>                           |
| Blisters                     | <ul style="list-style-type: none"><li>• Painful fluid filled lesion caused by excessive friction or pressure</li><li>• May burst due to friction or pressure from shoes</li></ul>                                               |
| Athlete's foot (tinea pedis) | <ul style="list-style-type: none"><li>• Fungal infection causing itching, scaling, redness, and small blisters around toes or soles of feet</li><li>• Flaky, cracked skin with possible blisters</li></ul>                      |



# DIFFERENTIAL DIAGNOSES

| CONDITION               | PRESENTATION                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingrown toenail         | <ul style="list-style-type: none"><li>• Pain, tenderness, redness, and possible inflammation around toe</li><li>• May lead to infection if untreated</li></ul>                                                                                                                                                                                                                                  |
| Bruise                  | <ul style="list-style-type: none"><li>• Tender or swollen area with possible reddish or purpling colour</li><li>• Most often from recent trauma or injury</li></ul>                                                                                                                                                                                                                             |
| Poroma                  | <ul style="list-style-type: none"><li>• Benign growths that can appear anywhere but are more common on extremities (palms of hands, soles of feet). Growth occurs slowly over many years and is not bothersome/symptomatic</li></ul>                                                                                                                                                            |
| Skin carcinoma/melanoma | <ul style="list-style-type: none"><li>• <i>Basal cell carcinoma</i> - pearly white bumps or patches that may ooze or crust and look like an open sore</li><li>• <i>Squamous cell carcinoma</i> - small scaly bump or plaque, which may appear inflamed</li><li>• <i>Malignant melanoma</i> - small pink, red, or brown-black spot or bump with irregular borders and an uneven colour</li></ul> |



# WHEN TO REFER

## SITUATIONS REQUIRING PATIENT REFERRAL



- Lesion becomes red, inflamed, and/or purulent
- Systemic symptoms of infection (fever, malaise, nausea and vomiting)
- Immunocompromised, diabetes, peripheral vascular disease, >65 years of age or malnourished
- No symptom improvement after 10 to 14 days of salicylic acid treatment

## Quiz

Click the **Quiz** button to edit this object

### ✓ CHECK YOUR UNDERSTANDING

Select the correct answers to complete the following statements:

Calluses are typically found on the  of the foot while corns typically present on the  of the foot. A possible risk factor for developing calluses or corns includes  fitting of footwear and abnormal gait. The presence of calluses or corns in a patient who is diabetic warrants  by the pharmacist.



# ASSESSMENT

- Often based on appearance, symptoms, and location
- Absence of bleeding when layers are excised
  - Pinpoint bleeding is indicative of a plantar wart
- Rule out of other causes of thickened skin
  - Bunion, wart, or blister
- Medical imaging to identify if physical abnormality is a causative factor



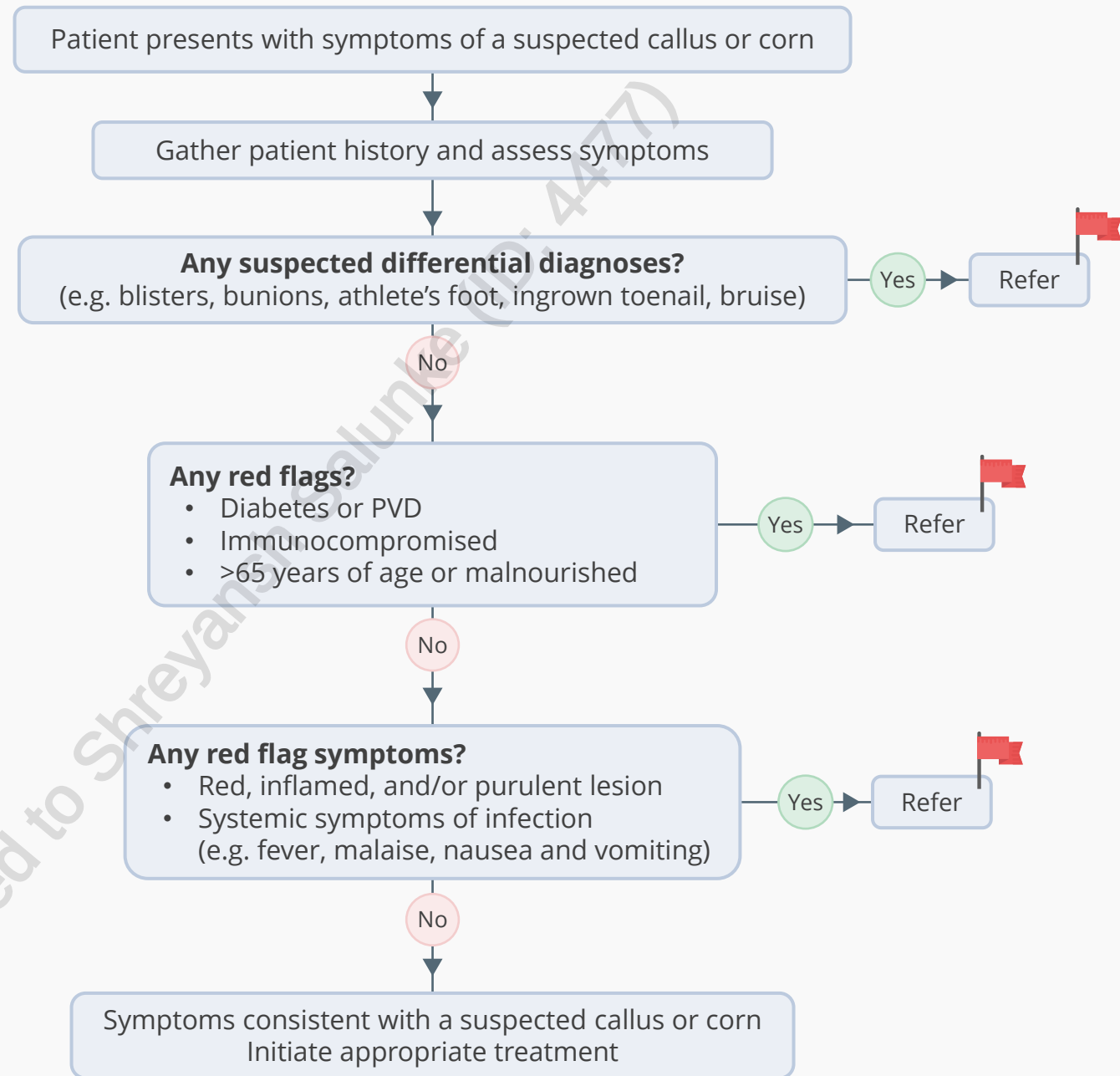


# ASSESSMENT

- Wherever possible, observe the lesion to differentiate between corn, callus, and other foot conditions
- Use SCHOLAR/HAMS to gather relevant history and risk factors
- Assess the symptom severity, onset, and other factors of the foot lesion
  - Acute or chronic? Frequency? Previous treatment? Characteristics and appearance?
- Ask about foot hygiene
  - Shoe/sock fitting, regular washing, avoiding bare feet
- Identify any red flags and refer promptly to physician
  - Purulence, dark-coloured lesions, treatment failure after 10 to 14 days of therapy, severe pain

# ASSESSMENT ALGORITHM

## CALLUSES AND CORNS



### LEGEND

PVD: Peripheral vascular disease

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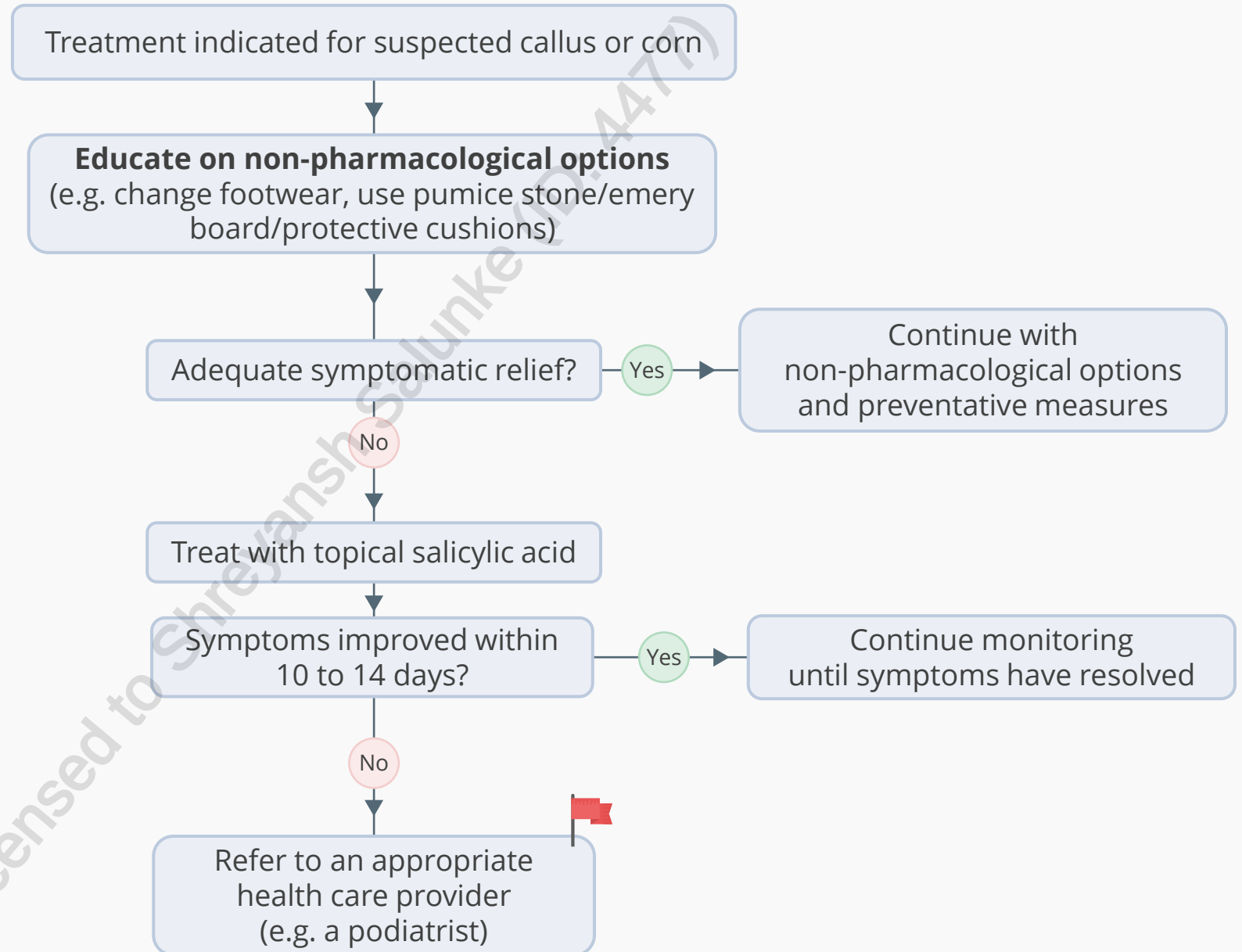
# TREATMENT



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# TREATMENT ALGORITHM

## CALLUSES AND CORNS





# NON-PHARMACOLOGICAL TREATMENT

- Changing footwear is often first-line for treatment of calluses or corns
  - Avoid tight-fitting shoes or footwear that has internal stitching or seams
- Use of a pumice stone or emery board may improve symptoms
  - Soak affected areas in water for 10 minutes before using a pumice stone and let feet dry before using an emery board
- Protective cushions or felt pads may stabilize a callus or corn
  - Reduce symptoms of pain and irritation
- Maintaining moisture and removing dead skin is a preventative measure to reduce further lesions
  - Avoid the use of moisturizers between the toes as this may cause excessive softening
  - Moisturizers with urea aid in removing dead skin but may cause burning or irritation on dry and cracked skin



# PHARMACOLOGICAL TREATMENT OPTIONS

| MEDICATION                                   | ADMINISTRATION                                                                                                                                                                                                                  | ADVERSE EFFECTS AND CONSIDERATIONS                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Salicylic acid</b><br>(keratolytic agent) | <ul style="list-style-type: none"><li>• 17.6% liquid</li><li>• Apply once or twice daily as needed for 10 to 14 days</li></ul>                                                                                                  | <ul style="list-style-type: none"><li>• Side effects: excessive burning or irritation</li><li>• Avoid on broken, irritated, or infected skin</li><li>• Avoid in PVD, impaired circulation, and diabetics</li><li>• Liquid formulation is highly flammable</li><li>• Apply pads or plasters to clean, dry surface and tape into place if necessary</li><li>• Pads or plasters may adhere better than liquid formulations</li></ul> |
| <b>Salicylic acid</b><br>(keratolytic agent) | <ul style="list-style-type: none"><li>• 40% pads</li><li>• Apply a new pad to the affected area every 2 days and repeat for 10 to 14 days</li><li>• Ensure inner circle of pad is placed directly on a corn or callus</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Quiz

Click the **Quiz** button to edit this object

### CHECK YOUR UNDERSTANDING

Is the following statement true or false?

Application of topical salicylic acid may cause a burning sensation and irritation to the skin.

☐ True

☐ False

## Quiz

Click the **Quiz** button to edit this object

### CHECK YOUR UNDERSTANDING

Type in the space provided the time frame you think would be appropriate for follow up with the patient after prescribing treatment for calluses and corns.



# MONITORING

- Monitor calluses and corns daily until healing is complete
- Clinical improvement should be seen within 10 to 14 days of treatment
- Monitor for adverse effects of salicylic acid such as excessive burning or irritation
- If skin around a corn or callus is broken or cracked, avoid use of salicylic acid until healing is complete



# CLINICAL PEARLS

- Promote good foot hygiene and ensure shoes are well-fitted
- Regularly inspect areas prone to calluses and corns e.g. inspect feet daily
  - Especially in diabetics, PVD, immunocompromised
- Strategies for preventing future corns or calluses
  - Moisturizers, protective padding, insoles
- Salicylic acid pads and plasters adhere to lesions better than liquid
  - Need to change pads/plasters every other day and remove dead skin
- Podiatrist referral for:
  - Persistent or severe symptoms
  - Immunocompromised
  - PVD or diabetics

# PATIENT CASE





# CASE BACKGROUND

**Background:** JD is a 27-year-old male who presents to your clinic with what he states to be a wart on top of his large toe. JD states he has had this lesion for a few weeks. JD also states he finds it sometimes painful to walk but not necessarily painful to manipulate. JD wants to know what products he can use to try to remove this lesion. He is open to using non-pharmacological or pharmacological treatments. You observe the lesion as a tough buildup of skin with a hardened centre.



## PAST MEDICAL HISTORY

- Seasonal allergies

## ALLERGIES

- No known drug allergies

## SOCIAL HISTORY

- No smoking
- Daily exercise
- 3 to 4 alcoholic drinks/week

## MEDICATIONS

- Loratadine 10 mg once daily, as needed



# CASE QUESTIONS

1. Which ONE of the following statements regarding foot lesions is true?
  - a) Calluses are painful to walk on whereas corns are not
  - b) Corns are self-limiting whereas calluses are not
  - c) Warts and calluses are considered an infection whereas corns are not
  - d) A centre core of hardened skin indicates a corn
  
2. Which type of lesion does JD have?
  - a) Callus
  - b) Corn
  - c) Wart
  - d) Blister



# CASE QUESTIONS

3. Which ONE of the following non-pharmacological treatments can be recommended to JD?
  - a) Tight-fitting shoes reduce foot slippage and possible calluses
  - b) Use moisturizers on entire surfaces of feet to prevent calluses and blisters
  - c) Cover the area with essential oils such as sage oil to eliminate dry skin
  - d) Soak feet in water and scrape lesion with a pumice stone to remove dead skin
  
4. Which ONE of the following statements regarding salicylic acid administration is FALSE?
  - a) Apply the liquid to the corn and surrounding skin to prevent spreading
  - b) Trim the pad/plaster to fit the area of the lesion and tape it in place if necessary
  - c) Avoid use of salicylic acid on broken skin or cracked skin
  - d) Apply a new pad or plaster every other day for up to 14 days



# CASE SOLUTIONS

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# REFERENCES

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